

MLDM Assignment 11

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 11

This are your test results

Knowledge Questions

 Context dependent classification

Status

Seen but not answered

Your score

0 / 6

0%

Response

Select 'Right/Wrong' for each statement.

Note:

- Correct answer: 1.5 point
- Incorrect answer: -0.5 point
- Unanswered: 0 points
- Max score: 6 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	In Context-Dependent Classification, the class to which a feature vector is assigned is independent of the values of other feature vectors.
☐	☐	☐	The sequence or the order of the feature vector is important in Context-Dependent Classification.
☐	☐	☐	The total number of possible class sequences is dependant on the number of classes and on the number of feature vectors that are present in the dataset.
☐	☐	☐	In Context-Independent Classification, there exists a relationship among the classes.

 Markov Chain model

Status

Not started

Your score

0 / 6

0%

Response

Select 'Right/Wrong' for each statement.

Note:

- Correct answer: 1.5 point
- Incorrect answer: -0.5 point
- Unanswered: 0 points
- Max score: 6 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	The probability density function in one class does not depend on the other class.
☐	☐	☐	In a first-order Markov model, the class dependence is limited only within two successive classes.
☐	☐	☐	For a given sequence of classes, the observations are statistically independent.
☐	☐	☐	Markov chain is a non-stochastic process.

☰

Hidden Markov Model

Status

Not started

Your score

0 / 6

0%

Response

A Hidden Markov Model has H hidden states and O observed states. Select 'Right/Wrong' for each statement.

Note:

- Correct answer: 1.5 point
- Incorrect answer: -0.5 point
- Unanswered: 0 points
- Max score: 6 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	The dimensions of the transition probability matrix of the HMM is $H * O$.
☐	☐	☐	The dimensions of the emission probability matrix of the HMM is $O * O$.
☐	☐	☐	The dimensions of the initial probability matrix of the HMM is $H * 1$.
☐	☐	☐	The sum of all the elements of the transition probability matrix of the HMM is equal to H.

☰

Essay

Status

Not started

Your score

0 / 12

0%

Response

Explain in your own words:

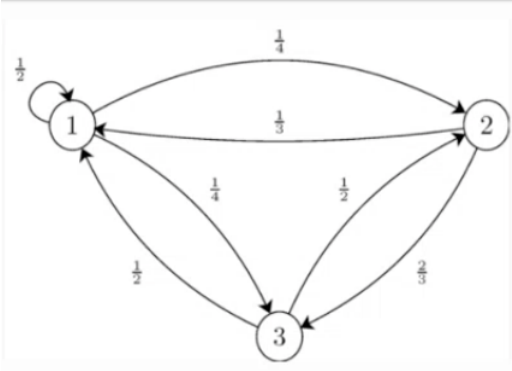
a.Transition Probabilities (6 Points)

b.Emission Probabilities (6 Points)

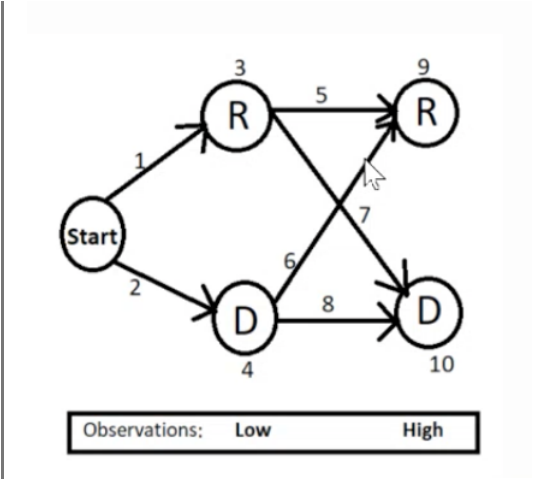
0 word

🧩 Practice

For the transition probabilities question, consider the following image:



For the Viterbi Algorithm question, consider the following image:



Transition probabilities

Status

Not started

Your score

0 / 15

0%

Response

Consider the state transition diagram of a Markov chain model with three states $S=\{1,2,3\}$ as shown in the figure and give your answers to the questions asked below with two digits after the decimal point.



fig:state-transition diagram

1. The sum of the transition probabilities from state 3 to two other states =

2. The initial probability of state 1 is $\frac{1}{4}$ and the initial probability of state 2 is $\frac{1}{4}$. Then the initial probability of state 3 =

3. The probability of transitioning from state 3 to state 2 and then to state 1 is =

Hidden states

Status

Not started

Your score

0 / 5

0%

Response

Suppose that there is an input sentence "I like to watch a movie". The possible part of speech tags that each of the words in the sentence can take is given below. The parts of speech tags are considered as the hidden states.

(PRP: pronoun, VB: verb, NN: Noun, DT: Determinant, TO: to)

- I: PRP
- like: VB
- to: TO
- watch: NN, VB
- a: DT
- movie: NN

The possible hidden state sequences for the above sentence is =

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3/5

 Viterbi algorithm

Status Not started

Your score 0 / 18 0%

Response

We want to estimate whether it rained or stayed dry over two days using a Hidden Markov Model. Thus, we observe the air pressure per day and record either *Low* or *High* air pressure. Initial probabilities are given as 0.6 for *Rain* and 0.4 for *Dry*. Below you find the emission probabilities and the transition probabilities required for your calculations:

Emission Probabilities	Low	High
Rain	0.6	0.4
Dry	0.3	0.7

Transition probabilities

$P(W_k = \text{Rain} \mid W_{k-1} = \text{Rain}) = 0.7$

$P(W_k = \text{Dry} \mid W_{k-1} = \text{Rain}) = 0.3$

$P(W_k = \text{Rain} \mid W_{k-1} = \text{Dry}) = 0.2$

$P(W_k = \text{Dry} \mid W_{k-1} = \text{Rain}) = 0.8$



Given is a sequence of observations: [*Low*, *High*]. Apply the Viterbi algorithm to compute the cost of every state at every stage. Please index your computations with the indices from the graph figure above.

Important: do **not** use log probabilities for the scores. provide your answer up to all the digits after the decimal point.

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

Match

Status

Not started

Your score

0 / 2

0%

Response

Based on the computations performed in the above task, Decide what is the sequence of hidden states that matches the best the given sequence of observations.

	Dry	Rain
State 1	☐	☐
State 2	☐	☐

Programming

Viterbi algorithm

Status

Not started

Your score

0 / 30

0%

Response

As a part of the programming task, implement the Viterbi algorithm and to ensure the correctness of the program, output the hidden states of the observed sequence [sunny, rainy] with the transition and emission probabilities given below. The initial probability of being Happy and Sad is 0.1 and 0.9 respectively.

Table 1: Emission Probabilities

	sunny	rainy
Happy	0.8	0.2
Sad	0.4	0.6

Transition Probabilities

$p(w_k = \text{Happy} \mid w_{k_1} = \text{Happy}) = 0.7$

$p(w_k = \text{Sad} \mid w_{k_1} = \text{Happy}) = 0.3$

$p(w_k = \text{Happy} \mid w_{k_1} = \text{Sad}) = 0.4$

$p(w_k = \text{Sad} \mid w_{k_1} = \text{Happy}) = 0.6$

Note: Please upload the .ipynb file and make sure to print the output.

File

No file uploaded